

23AHS21 - probability & statistics

Unit-III : probability Distributions :

Assignment - III

- 1) A die is thrown 6 times. If getting an even number is a success, find the probabilities of
 - i) at least one success.
 - ii) at most three successes. [CO3: Apply]
 - iii) 4 successes.
- 2) If 2% of light bulbs are defective, Find
 - i) At least one is defective
 - ii) Exactly 7 are defective
 - iii) $P(1 < x < 8)$ in a sample of 100. [CO3: Apply]
- 3) Find the mean and standard deviation of a normal distribution in which 7% of items are under 35 and 89% are under 63. [CO3: Apply]
- 4) State and prove Chebyshev's inequality. [CO3: Understand]
- 5) Find the probability that out of 100 patients between 84 and 95 will survive a heart-operation given that the chances of survival is 0.9. [CO3: Apply]